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Kelas : PSTI-D (PSD-E)

* binary Interpolation

arr E : [5, 12, 19, 23, 31, 37, 45, 52, 68, 74, 89, 95]

target : 37

> Versi double

Iterasi 1

$$pos = low + \left[\frac{target - arr[low]}{arr[high] - arr[low]} \times (high - low) \right]$$

$$pos = 0 + \left[\frac{37 - 5}{95 - 5} \right] \times (11 - 0)$$

$$= 0 + \left[\frac{32}{90} \right] \times 11$$

$$= 0 + 3,91$$

$$= 3$$

arr[3] = 23

low = pos + 1

$$= 3 + 1 = 4$$

Iterasi 2 :

low = 4

$$pos = 4 + \left[\frac{37 - 31}{95 - 31} \right] \times (11 - 4)$$

$$= 4 + \left[\frac{6}{64} \right] \times 7$$

$$= 4 + ~~0,09375~~ 0,09375 \times 7$$

$$= ~~4,06625~~ 4 + 0,09375 = 4,09375$$

$$pos = 4$$

arr[4] = 31

$$low = 4 + 1 = 5$$

Iterasi 3

low = 5

$$pos = 5 + \left[\frac{37 - 37}{95 - 37} \right] \times (11 - 5)$$

$$= 5 + \left[\frac{0}{58} \right] \times 6$$

$$= 5 + 0 \times 6$$

$$= 5 + 0$$

$$= 5$$

$$arr[5] = 37$$

target 37 =

$$low[5] = target[5]$$

$$37 = 37$$

berkamu

Vari: int

Iterasi 1

$$\text{pos} = \text{low} + \left[\frac{\text{target} - \text{arr}[\text{low}]}{\text{arr}[\text{high}] - \text{arr}[\text{low}]} \right] \times (\text{high} - \text{low})$$

target: 37

$$\text{pos} = 0 + \left[\frac{37 - 5}{95 - 5} \right] \times (11 - 0)$$

$$= 0 + \left[\frac{32}{90} \right] \times 11$$

$$= 0 + 0.355 \times 11$$

$$= [3.91]$$

$$\text{pos} = 3$$

$$\text{arr}[3] = 23$$

target > pos

$$\text{low} = \text{pos} + 1$$

$$= 3 + 1$$

$$= 4$$

Iterasi 2

$$\text{low} = 4$$

$$\text{pos} = 4 + \left[\frac{37 - 81}{95 - 81} \right] \times (11 - 4)$$

$$= 4 + \left[\frac{6}{14} \right] \times 7$$

$$= 4 + 0.4285 \times 7$$

$$= 4 + 3 = 7$$

$$\text{pos} = 7$$

$$\text{arr}[7] = 31$$

target > pos

$$37 > 31$$

$$\text{low} = \text{pos} + 1 = 7 + 1 = 8$$

Iterasi 3

$$\text{low} = 8$$

$$\text{pos} = 8 + \left[\frac{37 - 37}{95 - 37} \right] \times (11 - 8)$$

$$= 8 + \left[\frac{0}{58} \right] \times (3)$$

$$= 8 + 0 \times 3$$

$$= 8 + 0$$

$$= 8$$

$$\text{pos} = 8$$

$$\text{arr}[8] = 37$$

↳ ketemu